



Coal allocation optimization based on a hybrid residual prediction model with an improved genetic algorithm

Ming Liu^{a,b}, Ziqi Yu^a, Boran Li^a, Qingjie Wang^c, Huawei Ren^c, Dong Xu^{d,*}

^a Department of Applied Mathematics, School of Mathematics and Statistics Sciences, Changchun University of Technology, Changchun, 130012, PR China

^b Key Laboratory of Symbolic Computation and Knowledge Engineering of Ministry of Education, Jilin University, Changchun, 130012, PR China

^c China Risun Coal Chemicals Group Limited, Beijing, 100029, PR China

^d Department of Electrical Engineering and Computer Science, Christopher S. Bond Life Sciences Center, University of Missouri, Columbia, MO, 65211, USA

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ABSTRACT

The objective of the coal blending optimization problem is to find an optimal coal blending in the feasible domain such that the blended coal meets the quality requirements at the end of the coking process and the cost of coal blending is minimized. This paper proposes a hybrid residual prediction model and an improved genetic algorithm to solve this problem and predict coke quality. For this purpose, a hybrid residual prediction model is used to predict coke quality. The model first uses a random forest feature extraction method to reduce the dimensionality of the data, and then trains several prediction models such as eXtreme Gradient Boosting (XGBoost), Adaboost and Light Gradient-Boosting Machine (lightGBM) for different coke indicators an improved genetic algorithm based on the adaptive weighted genetic algorithm (awGA) and another improved genetic algorithm based on a priori knowledge and adaptive random initialization method were designed and implemented to solve the optimization problem under strict constraints (P-awGA). The experimental results show that using the hybrid residual prediction model and the improved genetic algorithm can accurately predict the coke quality and use less time to obtain a lower-cost coal blending solution.

1. Introduction

Studies show that in recent years coal has been the main energy source, consuming 29% of global fossil energy demand, generating 41% of the world's total electricity, and accounting for 44% of world industrial demand ("Data & Statistics - IEA," n.d.; Wu and Chen, 2018). Coal's massive consumption as an industrial raw material and energy source worldwide has caused widespread concerns in environmental pollution, and cost of fossil energy (Jaramillo and Muller, 2016; Shon et al., 2020; Xie et al., 2020; Zhang et al., 2020). Coke, an important coal component accounts for approximately 19% of total coal production worldwide, with approximately 650,000 kiloton of coke being produced each year ("Data & Statistics - IEA," n.d.). The relatively high cost of good quality coal as an ideal raw material for coke and its low global production has forced the coking industry to use poor quality coal as an alternative for production. However, poor quality coal has more impurities and emits a large number of pollutants during the coking and use processes, including volatile organic compounds, such as alkanes, olefins and alkynes, as well as fine particulate matter fine-tune, carbon

monoxide, carbon dioxide, nitrogen oxides and sulphur dioxide. These pollutants not only have a significant impact on the atmosphere and people's health, but also reduce productivity and cause safety accidents (Gong et al., 2017; Guo et al., 2018; Tian et al., 2018; Zhang et al., 2020). These pollutants not only have a significant impact on the atmosphere and people's health, but also reduce productivity and cause safety accidents. Therefore, this paper explores a method to predict coke quality and optimize coal blending to minimize production costs and reduce resource wastes and environmental pollution while maintaining the quality of the coke produced. A prediction and optimization model based on machine learning and genetic algorithm for coal blending and coking is proposed for the actual coal blending and coking production process, as shown in Fig. 1.

In the coking process, as shown in Fig. 1, individual coals need to be obtained, grounded and mixed according to the corresponding ratios. The mixed coals are then put into the coke oven for a period of high-temperature coking to finally form coke. As the coals are mixed and the heated, the form, the surface and the chemical composition of the coals are changed (Chen et al., 2020; Guo et al., 2020; Wall et al., 2002;

* Corresponding author.

E-mail address: liuming@ccut.edu.cn (M. Liu).

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Xu et al., 2016). However, as the blended coal is made up of a mixture of coals, it is challenging to determine the overall quality of blended coals, taking into account the quality of the individual coals and the physical changes during the blending process (Díez et al., 2002). The change from blended coal to coke involves complex reactions among various components of the blended coals at high temperatures. Although the traditional method of using simplified blast furnace coking before assaying can be used on a trial basis to obtain quality indicators for the coke as a product (Shen and Yu, 2016), this has the disadvantage that actual coking tests are required to obtain a true quality specification. Throughout the coking process, individual coal ratios are usually given by professionals based on experience and expectations of the target coke quality. Actual coking tests are carried out several times and the ratios are constantly adjusted to obtain the final target coke. Obviously, these multiple tests inevitably result in higher production costs and waste of resources. The environmental pollution and waste of resources caused by the coking process and the use of coke is a severe obstacle to the development of the coking industry and its subsidiary industries (Martinson et al., 2017; Zhu et al., 2019). Therefore, it is necessary to develop a predictive model of coke quality and to study the optimization of the composition ratio of coking coal raw materials and utilize the advantages of each raw material to improve the quality of coke while reducing resource consumption and costs as well as protecting the environment (Jiao et al., 2018).

The selection of the coking coal feed is the first step in the coking chemical process and the ratio of the selected single coal feed is directly related to the yield and quality of the coke output from the coking process (Díez et al., 2002; Jiao et al., 2018; van Krevelen and Schuyer, 1957; Zhang et al., 2004). The coke has a high crushing strength (M_{40}), abrasion strength (M_{10}), impurity level (ash A_d , sulphur content $S_{t,d}$) and the grade parameters of the coke (fugitive constituent) V_{daf} (Li, 2016),

which directly affect its combustion and the effect of iron making in the blast furnace.

And the performance of coke mainly depends on the grade parameters, the level of impurities (ash A_d , Sulphur $S_{t,d}$), agglomeration characteristics, etc. (van Krevelen and Schuyer, 1957). Blending coal is formed by mixing single coals in a certain proportion, and its performance index depends on the quality and proportion of each single coal. As the physical changes (crushing, mixing, etc.) occur in the process of converting individual coals to blended coals, the quality of the individual coals can be approximated by weighted summation of the corresponding ratios (Zhang et al., 2004).

For coke quality prediction methods, there are now several major research directions in the industry as follows.

Approaches based on coal petrography and optical analysis model (Smędowski and Krzesińska, 2013; Flores et al., 2017; Jiao et al., 2018; Chen et al., 2020; Yuan et al., 2020) predict coke quality by observing changes in the microstructure and optical properties of blended coals and coke before and after coking. Approaches based on fluid dynamics model (Wall et al., 2002; Wałowski, 2019; Xing et al., 2019) analyze the chemical and physical changes that occur in the coking process by modelling the fluid motion of the gases in the coking process and the pulverized coal under heat treatment conditions, and then to predict the quality of the coke in combination with the data obtained from subsequent tests on the quality of the produced coke. Data mining algorithms such as random forests (Chehreh Chelgani et al., 2016), back propagation neural networks (Khoshjavan et al., 2011) and integrated deep learning (Yin et al., 2020) were developed to analyze the quality indicators of blended coal and coke, learn the features between the indicators and finally use the obtained feature vectors for regression to build a predictive model of coke quality. Although these approaches have some success, considering the heterogeneity of coal, i.e., coals of

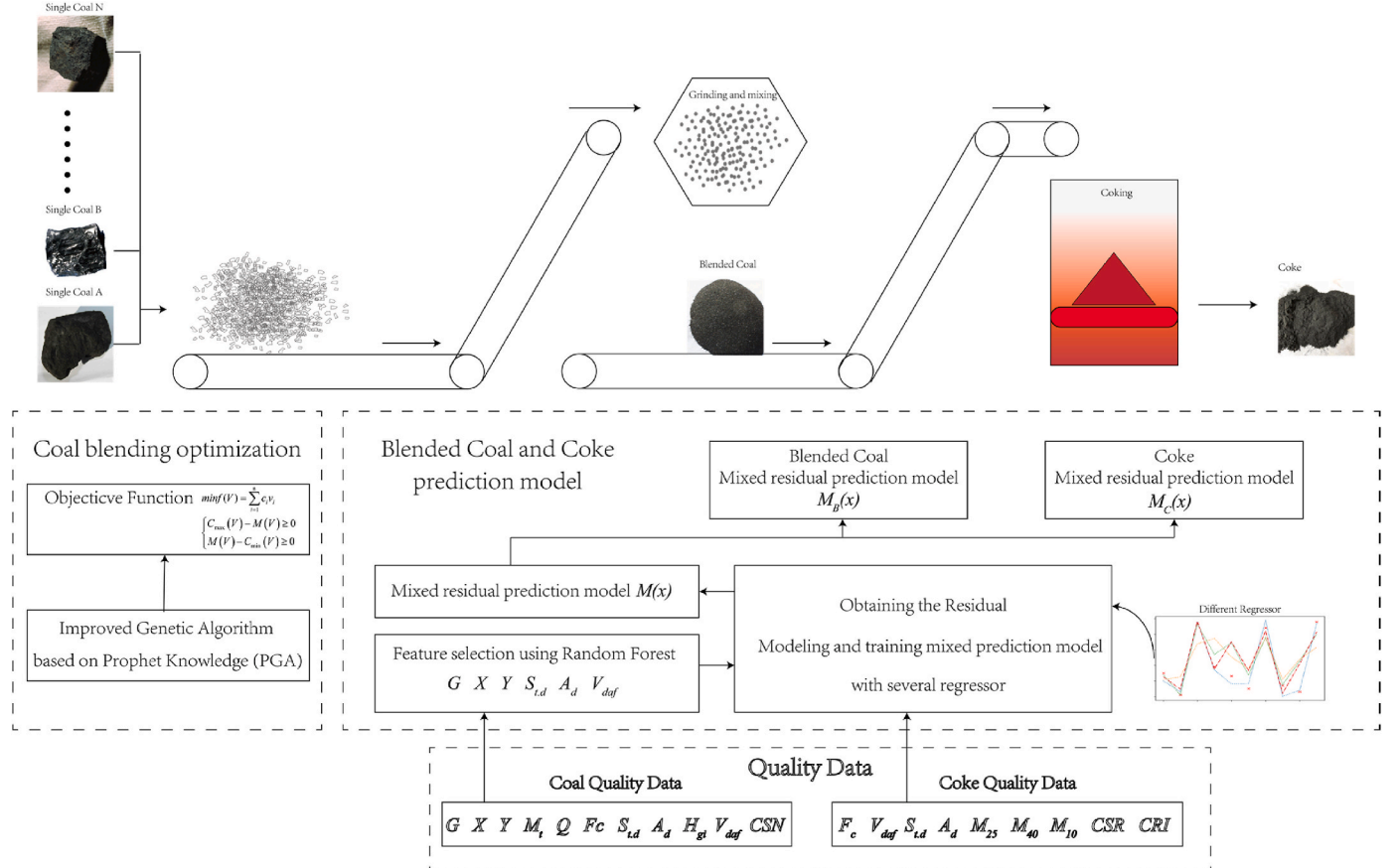


Fig. 1. Diagram of the coal preparation and coking process integrated with the predictive optimization model.

similar quality with different structures and compositions due to differences in geological conditions, and the fact that there are too many types of single coals as coke feedstock with different microstructures and qualities, it is difficult to predict coke quality from the properties of single coals and mixed coals themselves (Flores et al., 2017; Meng et al., 2017). Therefore, it is necessary to consider a new approach to combine the strengths of different models in order to enhance the reliability and effectiveness of coke quality prediction.

Furthermore, for coal blending optimization, practitioners want to optimize production costs while maintaining coke quality. A number of optimization methods have been proposed by the industry to solve such problems. These methods are usually based on a predictive model for matching coal and coke, as summarized above, constructing an objective function that reasonably describes the optimization problem, and then choosing a more appropriate optimization algorithm to solve the problem (Tian et al., 2016). In recent research, it has been found that, due to the ability to perform high-performance optimization with non-linear optimization under complex constraints, genetic algorithms (Chakraborty and Chakraborty, 2012; Ilbeigi et al., 2020; Kurnadi et al., 2022; Xi-Jin et al., 2009; You et al., 2020) and particle swarm algorithms (Jagtap et al., 2020; Li and Yao, 2017; Yuan et al., 2020) have been used in the industry to solve some complex optimization problems. Genetic algorithms have superior performance in solving complex constrained optimization, non-linear optimization problems and also have better robustness compared to other optimization algorithms. Genetic algorithms also allow the population size to be increased to improve the efficiency and robustness of the search, which inevitably increases the time and space overheads but can be reduced by parallel and distributed operations. (Akopov et al., 2019; Kim and Kim, 2017; H.-C. Lu et al., 2020). However, the above optimization algorithms still exhibit certain limitations: encountering difficulties in obtaining solutions that satisfy all constraints is highly probable in solving problems with complex constraints (Ma et al., 2021). Furthermore, traditional heuristic optimization algorithms (including genetic algorithms and particle swarm optimization) still demonstrate poor performance in terms of time efficiency (Abbasi et al., 2020). These issues have been confirmed to potentially be fatal in coal blending industrial production (Yuan et al., 2020). Therefore, proposing a novel optimization algorithm that can balance time efficiency and solution success rate is an urgent and significant challenge for researchers to address.

In summary, in order to optimize the cost of coal distribution, there are two key issues as follows.

1. Develop a predictive model for blended coal and coke quality with a high confidence level

Considering that the quality relationship between blended coal and coke is too complex, it is difficult to extract feature vectors that are more closely related to the corresponding quality indicators. Since the relationship between blended coal and coke is not simply linear, this leads us to need a reasonable non-linear model to describe the relationship.

2. Select a reasonable optimization algorithm to solve the optimization objective function

As the optimization of coke costs requires the use of a non-linear predictive model as described above, with a non-convex feasible domain, there are also complex and numerous input constraints and proportionality constraints to be considered in practice. This can make it difficult or costly to find an optimal solution if a suitable optimization algorithm is not found.

To address the above issues, this paper proposes a coal allocation optimization system based on a hybrid residual prediction model with a genetic algorithm, which is proven to accurately predict coke quality and effectively optimize coal allocation costs. In order to improve the robustness of the model and solve the problem of good model fit but

poor prediction results, we propose the use of residual prediction to reduce the influence of ' in the data set by predicting the difference between the target index and the input values, thus improving the data utilization and the prediction effectiveness of the model (Peng et al., 2015; Zhou et al., 2017). We then use a random forest algorithm to predict the residuals and extract the eigenvectors associated with the target metric, to minimize the effect of irrelevant variables to improve prediction accuracy and reduce the risk of overfitting (Chehreh Chelgani et al., 2016). Finally, we tested Adaboost, lightGBM and XGBoost and selected the method with the highest prediction accuracy to build the prediction model, and mixed the prediction models built by the different methods to improve the prediction performance.

After building a coke prediction model with high confidence, we propose an improved genetic algorithm for the coal allocation optimization problem. First, the improved genetic algorithm is based on the adaptive weight genetic algorithm (awGA), which has good convergence and solution set distribution for optimization problems with complex constraints and high-dimensional decision variables, and is therefore suitable for solving industrial-scale optimization problems (Gu and Wang, 2020; Liu et al., 2020). The algorithm is therefore suitable for solving larger-scale optimization problems in industry. An improved genetic algorithm based on prior knowledge and adaptive stochastic initialization is then designed and implemented for solving optimization problems with strict constraints in order to provide better generalization capabilities. Finally, using the improved genetic algorithm, we successfully solved the optimal single coal proportioning solution and its required cost.

In summary, the main contributions proposed in this paper can be summarized as follows:

- 1) Presented an improved hybrid residual prediction model applied to the task of quality prediction for coke and blended coal. Building upon the Random Forest feature extraction method, the residuals between blended coal and single coal, as well as between coke and blended coal, are computed. Various prediction models are then employed to forecast these residuals. Finally, an optimal strategy for model selection is determined by evaluating metrics, selecting the model with the best predictive performance for different coal quality indicators, and integrating them into a comprehensive quality prediction model.
- 2) Introduced an enhanced prior knowledge-based and adaptive random initialization genetic optimization algorithm (P-awGA) with adaptive weights. Building upon the adaptive weight genetic algorithm (awGA), to address the challenge of generating initial populations under strict constraints, this paper employs an adaptive constraint initialization method and prior knowledge to generate the initial population. Furthermore, an adaptive weight adjustment method is employed to guide changes in constraint conditions, aiming to improve the success rate of solving and optimization effectiveness.
- 3) Proposed a novel coal allocation optimization method integrating the aforementioned algorithms. Utilizing past blended coal ratios as prior knowledge, a population is generated around them. The fitness of the population is calculated using the hybrid residual prediction model, and an adaptive search mechanism ensures that the generated population satisfies constraint conditions. Ultimately, the optimal individual is iteratively determined, with its corresponding single coal ratio serving as the result. Through validation with real coal blending data, the method proposed in this paper demonstrates superior success rates in solving under strict constraints and significantly reduces coal blending costs.

2. Method

2.1. Model framework

2.1.1. Hybrid residual prediction model

Firstly, this paper uses the linear weighting method to build the prediction model from single coal to blended coal. After multiplying the input single coal index data by the single coal ratio and summing them, the index data of blended coal is obtained.

After building a quality prediction model from single coal to blended coal, a hybrid residual prediction model from blended coal to coke is then built. The prediction models for each metric used in this paper are independent of the prediction models for the other metrics, so that the models are trained separately and do not affect each other. In training the models, we take the blended coal indicator G values, the X value, Y value, ash, volatile fraction, moisture, fineness and sulphur as feature vectors of the prediction model input, and the coke indexes M_{40} , ash M_{10} , ash A_d , Sulphur $S_{t,d}$, volatile matter V_{daf} , reactivity CRI , after coke reaction intensity CSR as the predicted target output. Because that the ash, volatile and sulphur targets of the coke are mainly dependent on the same type of targets of the blended coal (van Krevelen and Schuyer, 1957). The only coke indicators that can be predicted using residuals are the ash, volatile matter and sulphur content of coke. Therefore, in the prediction model of coke, only the residuals obtained from the difference between the above three indicators and the same indicators in the blended coal are used as the prediction target output for the training of the residual prediction model. While the other predictors, such as M_{40} , M_{10} etc., are tested and the prediction model with the best results is selected for training in this paper. Based on the residual prediction model, a random forest (RF) based feature extraction method is used to evaluate the importance of each feature using a Gini impurity calculation with 100 decision trees (Izquierdo-Verdiguier and Zurita-Milla, 2020). The features were ranked in descending order of importance. Afterwards, the indicators with the highest importance for the prediction target were selected based on the proportion of importance, used as features and tested using different regression methods. Then, the regressions were evaluated based on two regression indicators, the mean squared error (MSE) and the coefficient of determination (R^2), to select the better forecasting method for modelling the forecasting target. Finally, the models developed by using the above methods for predicting different coke indicators were combined to form a hybrid residual forecasting model.

Considering that the coking process from mixed coal to coke is a complex chemical change, its quality change is difficult to be fitted by the traditional linear regression model (Lu J et al., 2020). In this paper, after a comparative analysis of different non-linear prediction models, machine learning models such as the Support Vector Machine regression model, Adaboost regression model, XGBoost regression model, Bagging integrated regression model and LightGBM regression model were explored for hybrid integration.

The metrics used in this paper to evaluate the performance of the forecasting model are MSE and R^2 , which are mathematically defined as follows.

$$MSE = \frac{1}{n} \sum_{i=1}^n \left(Y_{True}^{(i)} - Y_{Pred}^{(i)} \right)^2, \quad (1)$$

$$R^2 = 1 - \frac{\sum_i \left(Y_{True}^{(i)} - Y_{Pred}^{(i)} \right)^2}{\sum_i \left(\bar{Y} - Y_{Pred}^{(i)} \right)^2}, \quad (2)$$

where the better the model fits as the value of MSE converges to 0. n is the number of data sets, and Y_{Pred} is the set of predicted values obtained by the prediction model based on the feature vector prediction, and Y_{True} is the set of real data, $\bar{Y}$ is the mean value of the real data.

2.1.2. Improved adaptive weight GA based on priori knowledge and adaptive random initialization (P-awGA)

For the coal blending optimization problem, the main objective of the coal blending optimization problem addressed in this paper is to reduce the coal blending cost as much as possible while meeting the coke quality index constraints. At the same time, in the actual coal blending operation, we have to consider not only how to meet the coke quality index, but also the current coal type and its stock problem and coal type ratio that the coke making company has. Therefore, after taking into account the above constraints, the following coal blending optimization model is proposed in this paper.

Firstly, the objective of minimizing production costs needs to be satisfied, so the following objective function is available.

$$\min f(V) = \sum_{i=0}^n c_i v_i \quad (3)$$

where n dimensional vector V represents the decision variables of the optimization problem. Each element v_i in V represents the corresponding individual coal i of the coal allocation scheme. c_i represents the unit cost of the corresponding coal type under this coal allocation scheme. The types of constraints to be considered in a coal distribution task coke quality constraints, inventory constraints and coal structural constraints.

For the coke quality constraint, the cokemaker requires the target coke's indicators such as X , V_{daf} , $S_{t,d}$, A_d , F_c , M_{40} , M_{25} , M_{10} , CRI and CSR , need to be kept within a certain range. To do this, we use two variables D_{max} and D_{min} to represent the upper and lower bounds of each indicator, respectively. Let the coke quality predicted from the decision variable V be (V) . Let I represent the different quality indicators of coke. Then the following constraint function can be obtained.

$$D_{min} \leq Q_i(V) \leq D_{max}. \quad (4)$$

For the inventory constraint, it is necessary to take into account the daily amount of coal to be consumed in the production plan of the coke plant, the estimated number of days of use and the actual inventory data. Thus, let the total amount of coal consumed per day be M_d , the estimated number of days of use be T , then the total expected coal consumption M_t can be obtained as follows.

$$M_t = M_d \times T. \quad (5)$$

Let the proportion of each individual coal i be v_i , then the total consumption of the corresponding coal type is M_{ti} . The total consumption of the corresponding coal type is the amount of coal in the stockpile, which cannot exceed the stockpile of the corresponding coal type S_i , we have the following set of inequalities.

$$\begin{cases} M_{ti} = v_i \times M_t \\ M_{ti} \leq S_i \end{cases} \quad (6)$$

For the coal structure constraint, the proportion of individual coal types with different properties given by the coking plant needs to be met. And the proportion of all coal types must sum to 100%. Therefore, the following inequality is given.

$$\begin{cases} P_{K_{min}} \leq Q'(V_K) \leq P_{K_{max}} \\ O_{K_{min}} \leq \sum_{i=0}^n v_{K_i} \leq O_{K_{max}} \\ V = V_0 \cup V_1 \cup \dots \cup V_K \cup \dots \cup V_N \\ \sum_{k=0}^N \sum_{j=0}^n v_{K_j} = 100\% \end{cases} \quad (7)$$

Among them, here the set V denotes the set of all coal types included in the current coal allocation scheme, the n dimensional vector V_K denotes the set of coal types in V that satisfy the given K_{th} individual coal quality indicators within the range of $P_{K_{min}}$, and $P_{K_{max}}$ vector v_K represents V the set of coal ratios that satisfy the given K_{th} individual coal quality in-

dicators with the range of $P_{K_{\min}}$, and $P_{K_{\max}}$. Single coal index $Q(A)$ denotes the quality index of the coal type corresponding to the elements contained in the vector. $O_{K_{\min}}$ and $O_{K_{\max}}$ denote the range of quality indicators the upper and lower limits of the ratio sum that satisfy the given K_{th} individual coal quality indicators within the range of $P_{K_{\min}}$, and $P_{K_{\max}}$.

Based on the objective function and constraints as above, we can describe the coal allocation optimization problem as a single objective nonlinear optimization constrained problem as follows :

$$\begin{aligned} \min f(v) &= \sum_{i=0}^n c_i v_i \\ \text{s.t.} \left\{ \begin{array}{l} D_{\max} - Q(V) \geq 0 \\ Q(V) - D_{\min} \geq 0 \\ \frac{S_i}{M_i} - v_i \geq 0 \\ P_{K_{\max}} - Q'(V_K) \geq 0 \\ Q'(V_K) - P_{K_{\min}} \geq 0 \\ O_{K_{\max}} - \sum_{i=0}^n v_{k_i} \geq 0 \\ \sum_{i=0}^n v_{k_i} - O_{K_{\min}} \geq 0 \\ V_0 \cup V_1 \cup \dots \cup V_K \cup \dots \cup V_N = V \\ \sum_{k=0}^N \sum_{j=0}^n v_{k_j} - 1 = 0 \end{array} \right. \end{aligned} \quad (8)$$

Because the model has a large number of complex constraint functions and the problem involves a large number of dimensions of decision variables, it is difficult to find the optimal solution while satisfying the constraints. Therefore, we use the penalty function method to convert the constrained optimization problem into an unconstrained optimization problem. When using the penalty function method to solve constrained optimization problems, since the optimization objective is to minimize cost, we apply a great penalty to the objective function value of the solution that violates the constraint by σ , which makes the objective function value of the solution so large that it is removed from the set of optimal solutions.

In order to solve the above model using a genetic algorithm, each coal allocation scheme is abstracted in this paper as the genotype of an individual in the population, and the optimization objective, i.e., the cost of coal allocation, calculated through the coal allocation scheme, represents the phenotype of that individual. The specific algorithmic flow of the genetic algorithm is as follows (Deb and Jain, 2014).

First, during the initialization of the algorithm, a set of individuals of the population size is randomly generated within the specified range of values of the decision variables according to the pre-defined parameters N of population individuals, and this set of population individuals forms the initial population P_0 . Once the initial population has been obtained, the algorithm iterates until the termination conditions are met.

In the t_{th} generation, the algorithm randomly selects each individual on the basis of the current population P_t , and performs an evolutionary operation on the selected individuals. The specified operator is called to complete its corresponding chromosome crossover and mutation operation, which will eventually result in a new population with a population size of N . A new population with a population size of P_t . This population is then combined with the original population to produce a population size of $2N$. A population of Q_t . This will result in a new pop-

ulation of population size. For each individual in the population Q_t . The objective function is called for each individual in the population to calculate the fitness of each individual. On the basis of the new population Q_t , by using an undominated ranking based on Pareto dominance, the Q_t . The individuals in the population are grouped into several non-dominated strata. Finally, individuals are randomly selected into the next generation population based on predefined reference points that satisfy the following conditions P_{t+1} in.

1. All individuals before the critical layer of the Pareto sort;
2. The point in the critical layer associated with the reference point;
3. When no point is associated with a reference point, select the point that is closest to the reference point.

Also, in order to minimize the time complexity of the algorithm, the pseudo-random number method is used in this paper instead of the random selection method to select individuals into the next generation population if multiple individuals satisfy the conditions of class 2 and class 3.

Since traditional genetic algorithms use a random selection method when selecting individuals to participate in the evolutionary operation, this method will lead to the fact that the individuals chosen to participate in evolution will not necessarily be the ones with the best fitness, which may lead to the genetic algorithm being trapped in a local optimum dilemma. In order to solve such a problem, the improved genetic algorithm proposed in this paper uses a tournament selection strategy in selecting individuals to participate in evolution, and selects the better individuals in the population to participate in the evolution as far as possible. This selection strategy not only speeds up the convergence of the algorithm, but also enables the algorithm to escape from the local optimum trap. At the same time, in order to improve the algorithm's ability to search for the global optimal solution in a high-dimensional decision variable space, the simulated binary crossover (SBX) operator with a large exponential distribution and the polynomial variation operator are selected to perform chromosome crossover and variation in the process of evolution in this paper (Deb and Jain, 2014; Yi et al., 2020).

Considering that the objective function of the coal allocation optimization problem has more complex constraints, it is difficult to satisfy all the constraints at once during the optimization search process when the constraints are relatively strict. However, the traditional genetic algorithm may converge to the vicinity of the global optimal solution that does not satisfy the constraints within the value interval of the decision variables when each individual of the initialized population does not satisfy all the constraints, and eventually fails to obtain a feasible solution that satisfies the constraints. In practice, we notice that if a feasible solution can be obtained when initializing the population, it can assist the algorithm in finding an optimal solution within the feasible interval, so that the algorithm can obtain an optimal solution that satisfies the constraint rather than an infeasible global optimal solution. Therefore, we improve the genetic algorithm by introducing priori knowledge and adaptive random initialization methods into the improved genetic algorithm proposed in this paper. Firstly, the genetic algorithm is divided into two stages. In the first stage, priori knowledge and adaptive stochastic initialization are used to find a feasible solution: when the algorithm receives as input priori knowledge, a coal allocation scheme that is known to satisfy the constraints, will generate a new population around it to prepare for the second stage of the evolutionary process. Otherwise, the algorithm performs adaptive random initialization, performing multiple rounds of adaptive random search in the solution space for an indeterminate number of times until it obtains a solution that satisfies the constraint as a priori knowledge. After obtaining a set of populations based on the priori knowledge, the algorithm enters into the genetic algorithm optimization search process described above. Ultimately, the optimal solution satisfying the constraint is obtained at the end of the loop.

P-awGA based on priori knowledge and adaptive random initialization method proposed in this paper is summarized in Fig. 2. see(Fig. 3)

In order to quantify the performance of the optimization algorithm proposed in this paper and to facilitate comparison with other optimization algorithms, the average computation time, the number of successful solutions and the objective function value of the average solution (i.e., cost) obtained from 100 solutions of the same coal allocation task using different algorithms are used in this paper in order to measure the effectiveness of the algorithms, considering that coal companies have certain demands for time efficiency and stability of the solution set for coal blending optimization tasks.

3. Results and discussions

3.1. Simulation conditions

All simulations covered in this paper were conducted on an AMD Ryzen 7 2700, 3.3 GHz CPU, 16 GB RAM and 64-bit Windows 10 operating system. The programming language used was Python. The data used in this paper came from the assay data of a coal company for the blended coal and coke as its products from 2016 to 2020, and its total sample data were 2939 sets. In order to improve the accuracy and

generalization of the model, the order of the data in the dataset was disordered and the dataset was split into a training set and a test set in the ratio of 8:2 before the prediction model was trained in this paper. For the comparison between the different genetic algorithms used in the cost optimization task, in order to reflect fairly and accurately the performance relationship between the algorithms, the parameters of the algorithms used in this paper were unified into the same parameters according to the actual situation as shown in Table 2.

Due to the fact that the maximum cost in the historical data is less than 2000, the objective function in this study is set to have a maximum value less than 2000. However, considering factors such as expansion of production that may exist in actual industrial production, production costs may increase (Chien et al., 2022). To ensure the robustness of the algorithm, this study sets a penalty value five times the maximum value. Therefore, the penalty function penalty value σ of the actual value is set to 10,000.

3.2. Evaluation of hybrid residual prediction model

In the training of the coke quality prediction model, feature extraction was first carried out using random forest, and feature vectors with high correlation with the target indicators were selected according to

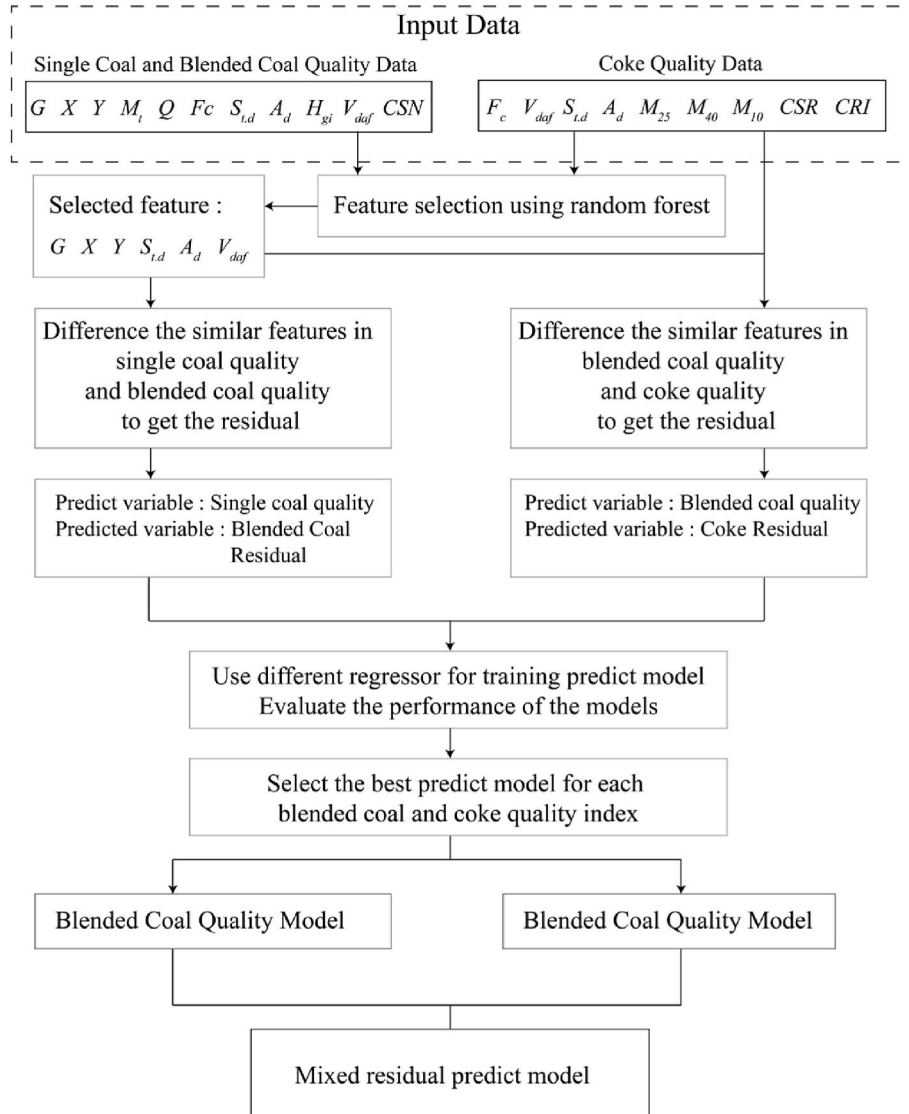


Fig. 2. Flowchart of hybrid residual prediction model.

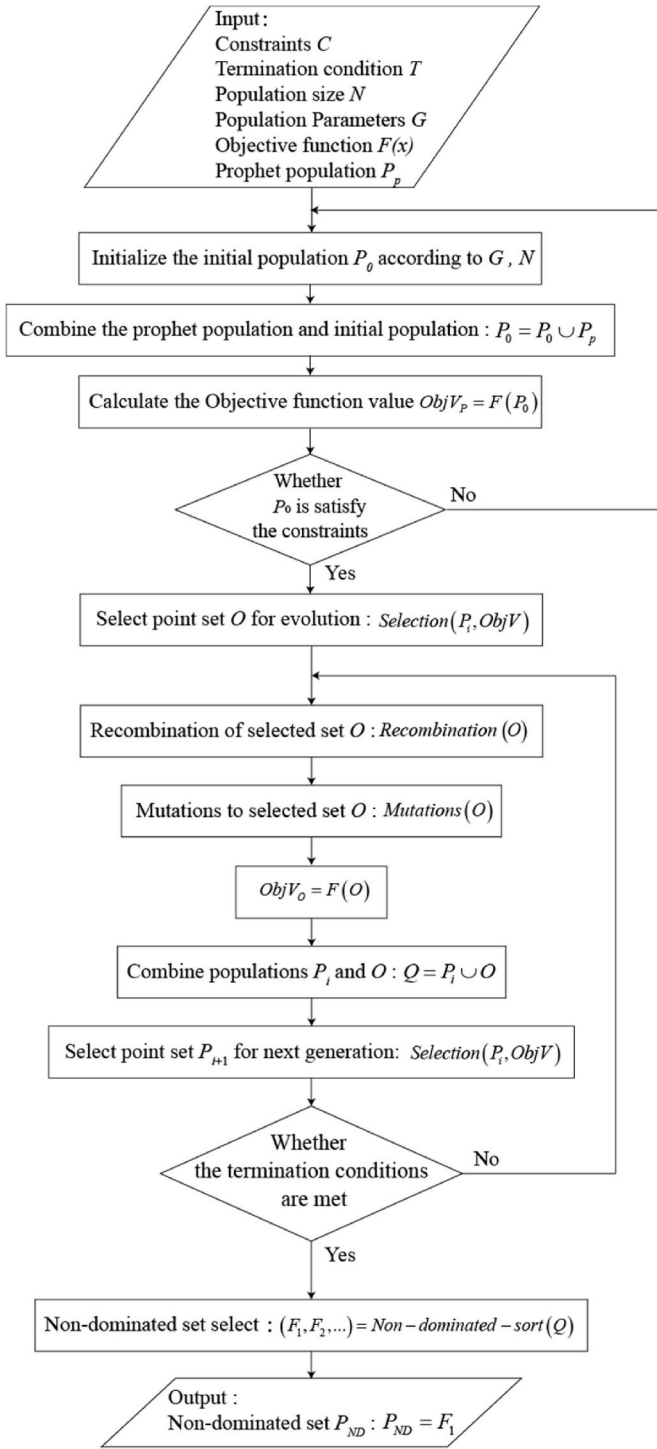


Fig. 3. Flowchart of an improved adaptive weight genetic algorithm based on a priori knowledge and adaptive random initialization methods.

the Gini impurity decay. Table 3 below shows the correlation between each coke indicator and the blended coal indicator.

As shown in Tables 3 and in this paper, after calculating the correlation between coke indicators and blended coal indicators, the indicators with correlation lower than 0.1 were excluded according to the correlation value of each indicator of blended coal corresponding to individual coke indicators, and the blended coal indicators with higher correlation used to build each coke indicator prediction model in this paper were bolded.

After obtaining the features with high correlation with the pre-

Table 1
Coke quality index.

	Name	Remarks	Lower bound	Upper bound	Type
V_{daf}	the grade parameters of the coke (fugitive constituent)	The organic substances in the coke are decomposed by heating to form a part of the liquid (in this case, the vapor state) and gaseous products with a small molecular weight, which account for the mass fraction of the coal sample.	0.9	1.8	Coke, Blended Coal
F_c	Fixed Carbon	Combustible solids in coke are the main components of coke combustion to generate heat	85	90	Coke
A_d	Impurity Level (Ash)	The content of various oxides in the solid residue of coke after ashing	12	13	Coke, Blended Coal
S_{td}	Impurity Level (Sulphur)	Sum of various forms of sulphur in coke	0.6	0.8	Coke, Blended Coal
M_{10}	Abrasion Strength	The ability of coke to resist external friction without producing surface glass to form chips or powders	4	7.5	Coke
M_{25}	Crushing Strength	One of the indicators of the ability of coke to resist external impact without breaking along cracks or imperfections in the structure	90	100	Coke
M_{40}	Crushing Strength	One of the indicators of the ability of coke to resist external impact without breaking along cracks or imperfections in the structure	80	100	Coke
G	Bonding Index	The performance index that coal particles can stick to each other into a block during the process of dry distillation	65	100	Coke, Blended Coal
X	Shrinkage Rate of Coke	Percentage of shrinkage of coke after reheating to higher than final temperature of coking to the original length	32	35	Coke, Blended Coal
Y	Thickness of Colloidal Matter Layer	Thickness of the colloidal layer formed continuously during the coking process of coal	21	28	Coke, Blended Coal
CSR	After coke reaction Intensity	The ability of the reacted coke to resist chipping and abrasion under	60	70	Coke

(continued on next page)

Table 1 (continued)

	Name	Remarks	Lower bound	Upper bound	Type
CRI	Reactivity	mechanical and thermal stress The ability of coke to resist breakage and wear when subjected to thermal stress and mechanical force at the same time as the temperature and atmosphere of the use environment	25	30	Coke

Table 2

Genetic algorithm test parameters.

Parameters	Values
Population size	100
Maximum number of iterations	100
Chromosome encoding type	RI
Upper and lower bounds of decision variables	[0,40]
Adaptive random initialization maximum number of times	100
Penalty function penalty value σ	10,000

Table 3

Correlation of various coke indicators with blended coal indicators.

Blending coal\Coke	V_{daf}	A_d	$S_{t,d}$	M_{40}	M_{10}	CSR	CRI
V_{daf}	0.37	0.08	0.15	0.18	0.26	0.26	0.22
A_d	0.23	0.71	0.12	0.25	0.17	0.27	0.23
$S_{t,d}$	0.12	0.07	0.53	0.18	0.18	0.15	0.20
G	0.12	0.05	0.08	0.25	0.16	0.15	0.15
X	0.12	0.03	0.08	0.13	0.12	0.09	0.11
Y	0.04	0.07	0.03	0.02	0.11	0.07	0.07

The blended coal indicators with higher correlation used to build each coke indicator prediction model in this paper were bolded.

dictors, we trained different prediction models using the selected feature vectors from the random forest to combine the advantages of different models to improve the prediction accuracy. We trained different models for each predictor separately, and to improve model reliability, conducted a ten-fold cross-validation to calculate the average MSE and R^2 to assess the effectiveness of these models. For clarity, the common residual model was Res-Model and the conventional non-residual prediction model using complete features was Model (where Model was replaced with the name of the respective model in practice) in this paper. Table 4 shows the prediction results of different prediction models for each quality indicator of coke, respectively, where the model corresponding to the bolded numbers is the final prediction model chosen for the

Table 4

Coke quality indicator residual prediction model prediction results.

	V_{daf}		A_d		$S_{t,d}$		M_{40}		M_{10}		CSR		CRI	
Regression Model	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2
RF-SVR	0.40	-0.17	0.02	0.73	0.09	-0.01	0.92	0.21	0.15	0.16	3.01	-0.14	3.70	-0.23
RF-Adaboost	0.24	0.30	0.01	0.81	0.04	0.53	0.31	0.73	0.11	0.41	1.10	0.57	1.22	0.59
RF-Bagging	0.09	0.72	0.01	0.79	0.02	0.72	0.01	0.98	0.02	0.85	0.56	0.78	0.63	0.78
RF-XGBoost	0.05	0.84	0.01	0.80	0.02	0.69	0.07	0.93	0.02	0.85	0.15	0.94	0.53	0.82
RF-lightGBM	0.05	0.83	0.01	0.82	0.03	0.60	0.11	0.89	0.01	0.90	0.43	0.83	0.38	0.87
SVR	0.36	0.01	0.03	0.69	0.09	-0.04	0.69	0.31	0.08	0.10	1.70	0.16	2.90	-0.01
Adaboost	0.17	0.52	0.02	0.74	0.04	0.52	0.23	0.76	0.06	0.26	0.83	0.59	0.95	0.66
Bagging	0.16	0.55	0.02	0.73	0.03	0.59	0.22	0.77	0.04	0.48	0.51	0.75	0.79	0.72
XGBoost	0.13	0.64	0.02	0.73	0.03	0.57	0.13	0.86	0.04	0.47	0.49	0.75	0.30	0.89
lightGBM	0.14	0.60	0.02	0.76	0.04	0.49	0.18	0.81	0.03	0.58	0.68	0.66	0.70	0.75

The model corresponding to the bolded numbers is the final prediction model chosen for the corresponding indicator in this paper.

corresponding indicator in this paper.

Tables 4 shows that the various improved coal allocation quality prediction methods that combine random forest feature extraction with residual prediction methods are significantly better than traditional prediction models. In this paper, the above tests are performed for each prediction target to select MSE and R^2 prediction models with better indicators, which are then combined into the hybrid residual prediction model proposed in this paper as shown in Fig. 4 below.

3.3. Evaluation of P-awGA improved genetic algorithm

In order to evaluate the effectiveness of the improved genetic algorithm proposed in this paper, the following optimization algorithms were used to compare with (P-awGA based on prior) knowledge and adaptive random initialization methods presented in this paper, as shown in Table 5.

Table 5 shows the average values of the metrics obtained from several tests of the above algorithms. It can be seen that P-awGA has a significantly higher chance of finding a feasible solution for a coal allocation task with tight constraints than several other algorithms, and is able to obtain a lower cost solution. While the traditional awGA has a significant advantage in terms of computational speed in a single iteration, it converges slowly and does not converge to near the global optimum in the same 100 iterations as the other algorithms. The P-awGA combines the advantages of the traditional awGA in terms of single iteration computation speed with the use of a priori knowledge to speed up the convergence and improve the accuracy of the solution, and preserve the a priori data when the solution set converges to the global optimum. At the same time, the P-awGA algorithm with adaptive stochastic initialization can maintain an acceptable solution rate in the absence of priori knowledge or when the priori knowledge does not satisfy the constraints. Therefore, the algorithm shows its great advantages in terms of efficiency and robustness.

3.4. Model validation

Based on the coal allocation prediction model and optimization algorithm presented above, this paper proposes a novel coal allocation optimization model. Here, the model will be applied to the optimization of the actual coal allocation scheme of the coal company. Firstly, as a comparison, this paper takes the original coal allocation scheme that was used and tested by the coal company as an example, as shown in Table 6 below. It can be seen that this coal blending scheme uses several individual coals to make coke and the cost of this scheme can be calculated to be \$993.96/t based on the price of the individual coals and the blending ratio. At the same time, according to Tables 7, we can get the quality index of the blending coal and coke made according to this coal blending scheme.

Based on the above constraints, the cost of coal allocation was optimized using P-awGA, resulting in a lower cost coal allocation

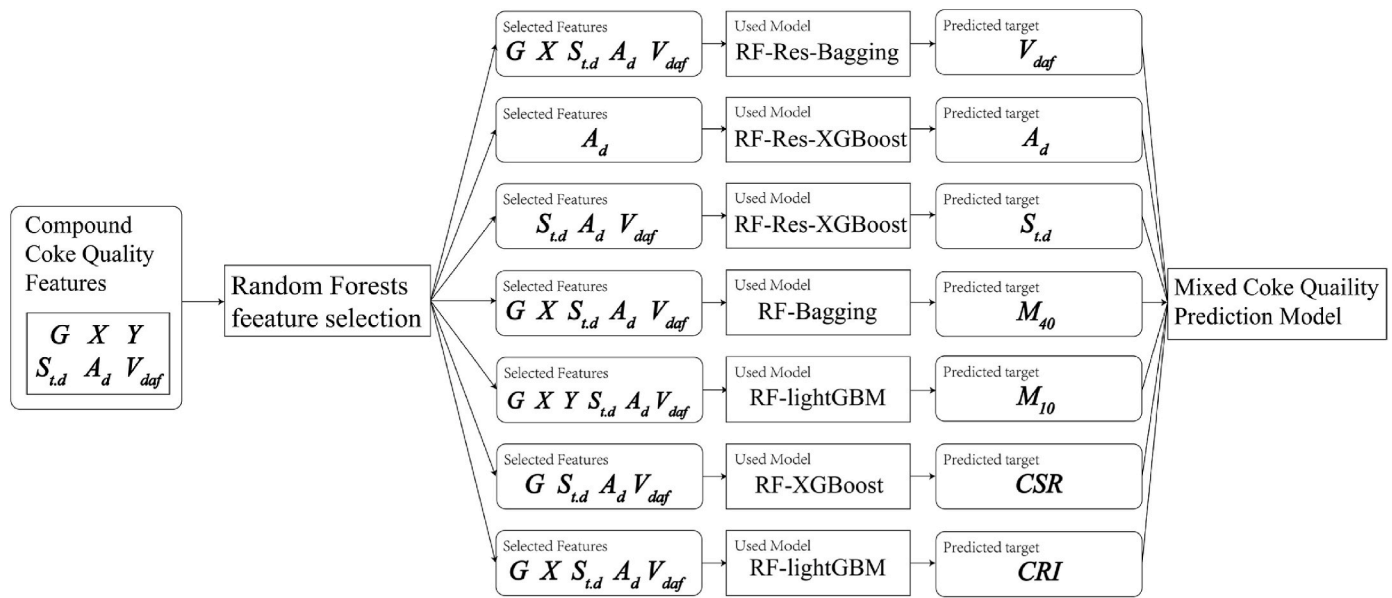


Fig. 4. Schematic diagram of the hybrid residual prediction model.

Table 5
Average performance metrics of different genetic algorithms for 100 solutions of the same coal allocation task.

Algorithm	Average calculation time (s)	Successful solutions	Average solution cost (CNY/t)
P-awGA	13.21	100	935.57
awGA	7.53	18	986.85
NSGA-III	15.03	16	961.39
NSGA-II	14.82	10	971.52

solution compared to the original coal allocation solution (see Table 6), while also ensuring that the coke quality was within the constraints (see Table 1). The cost of the optimized coal allocation scheme was calculated to be CNY928.33/t compared to CNY993.96/t for the original coal

allocation scheme, a total reduction of CNY65.63/t. It can be seen that the coal allocation optimization algorithm proposed in this paper can save approximately 6% of the cost in actual production. The company averages 5.9 million tons of coke per year and can save 387 million CNY by using this algorithm.

4. Conclusions

As the optimization of coal blending solutions is a challenging problem for the normal production and quality control of coke plants. However, the existing coal allocation and coke quality prediction methods based on traditional optimization algorithms and neural network model-based prediction algorithms have problems, such as low efficiency and poor prediction accuracy.

To address these problems, this paper proposes a coal allocation

Table 6
Comparison of historical coal allocation scheme and optimal coal allocation scheme obtained according to the optimization algorithm.

Coal Type	Historical coal ratio and costs		Optimized coal ratio and costs		Coal Data					
	Ratio	Costs (CNY/t)	Ratio	Costs (CNY/t)	V_{daf}	A_d	$S_{t,d}$	G	X	Y
1	0.10	849	0.21	849	13.66	10.53	0.37	0.86	0	0
2	0.19	858	0.35	858	37.76	8.52	0.52	78.44	47.53	12.32
3	0.25	1104	0.04	1104	36.90	9	0.45	88.97	39.73	19.38
4	0.08	761	0.16	761	38.13	8.92	1.96	94.14	43.73	25.18
5	0.17	1256	0.14	1256	22.83	9.92	0.61	77.73	28.50	20.01
6	0.10	1061	0.10	1061	30.03	12.16	1.69	89.12	32.77	19.90
7	0.11	814	0	814	14.85	10.12	1.28	15.95	0	0
Total (Historical)	1	993.96	0	0	28.90	9.70	0.81	70	41	13
Total (Optimized)	0	0	1	928.33	29.33	9.65	0.81	68.64	40.58	14.59

Table 7
Comparison of the weighted summation of the fit coal data and coke data with the real assay data based on the historical coal allocation list data.

Coal	V_{daf}	A_d	$S_{t,d}$	G	X	Y				
Predict Values	29.33	9.65	0.81	68.64	40.58	14.59	The upper part is the comparison between the predicted quality and the actual value of the matching coal, and the lower part is the comparison between the predicted quality and the actual value of the coke			
True Values	28.90	9.70	0.81	70	41	13				
Permissible error range	±1	±1	±0.1	±5	±1	±2				
Coke	V_{daf}	F_c	A_d	$S_{t,d}$	M_{25}	M_{40}	M_{10}	CSR	CRI	
Predict Values		1.01	86.25	12.83	0.71	94.90	88.48	4.664	62.33	26.88
True Values		1.00	86.46	12.50	0.70	94.20	87.80	4.60	61.10	28.10
Permissible error range		±0.1	±5	±1	±0.1	±5	±5	±0.1	±5	±5

optimization algorithm based on a hybrid residual prediction model and a genetic algorithm. Firstly, a random forest is used to extract features, and the feature vectors with high correlation with the target index are selected according to the decay of Gini's impurity. The filtered feature vectors are then used to train separate models for each predictor, and the MSE and R^2 to evaluate the effectiveness of these models. The trained models are then merged with models that predict other indicators, and the residual prediction method is used to address issues such as poor model robustness. The results of the simulation tests confirm that the prediction accuracy of the various improved coal distribution quality prediction models combined with the random forest feature extraction method proposed in this paper is significantly better than that of the traditional prediction models.

Then, based on the hybrid residual prediction model, this paper proposes an improved adaptive weight genetic algorithm based on prior knowledge and adaptive random initialization method. Based on the traditional genetic algorithm, in order to solve the problem that the initial population is difficult to generate under strict constraints, this paper uses the adaptive constraint initialization method and prior knowledge to generate the initial population. And the adaptive weight adjustment method is used to guide the changes in constraints to improve the success rate of solution and optimization effect. Finally, this paper demonstrates the advantages of the optimization algorithm proposed in this paper in terms of cost optimization and computational efficiency through practical examples.

In summary, the proposed hybrid residual prediction model and improved genetic algorithm can effectively predict and control the quality of coke in the actual production process, and can significantly reduce costs and resource consumption.

CRediT authorship contribution statement

Ming Liu: Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing. **Ziqi Yu:** Software, Visualization, Writing – original draft. **Boran Li:** Software, Validation, Writing – original draft. **Qingjie Wang:** Data curation, Formal analysis. **Huawei Ren:** Data curation, Formal analysis. **Dong Xu:** Funding acquisition, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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